

Calculus 2 - Homework 6 (Worked Solutions)

1.

Alternating Series Test: $b_n = 1/(2n - 1)$ is positive, decreasing, and $\lim b_n = 0$, so the series converges (this is the Leibniz series for $\pi/4$).
Absolute values: $\sum 1/(2n - 1)$; by limit comparison with $1/n$, $\lim [1/(2n-1)]/[1/n] = 1/2 > 0$, and $\sum 1/n$ diverges, so $\sum 1/(2n-1)$ diverges.
Converges but not absolutely $\rightarrow$ conditionally convergent.
ANSWER: conditionally convergent

2.

This classic reduction uses parts. Write $\sec^3 x = \sec x \cdot \sec^2 x$.
Let $u = \sec x$, $dv = \sec^2 x \, dx \rightarrow du = \sec x \tan x \, dx$, $v = \tan x$.
Let $I = \int \sec^3 x \, dx$.
 $I = \sec x \tan x - \int \tan x \cdot \sec x \tan x \, dx = \sec x \tan x - \int \sec x \tan^2 x \, dx$.
Use $\tan^2 x = \sec^2 x - 1$:
 $I = \sec x \tan x - \int \sec x (\sec^2 x - 1) \, dx = \sec x \tan x - \int \sec^3 x \, dx + \int \sec x \, dx$.
So $I = \sec x \tan x - I + \ln|\sec x + \tan x|$.
 $2I = \sec x \tan x + \ln|\sec x + \tan x|$, giving
 $I = (1/2)(\sec x \tan x + \ln|\sec x + \tan x|) + C$.
ANSWER: $(1/2)(\sec(x) \tan(x) + \ln|\sec(x) + \tan(x)|) + C$

3.

$\Delta x = 0.5$. Subinterval midpoints: 0.25, 0.75, 1.25, 1.75.
 $f(x) = x^2$ at midpoints: 0.0625, 0.5625, 1.5625, 3.0625.
Midpoint: $M = \Delta x \cdot (\text{sum of midpoint values})$
 $= 0.5 \cdot (0.0625 + 0.5625 + 1.5625 + 3.0625) = 0.5 \cdot 5.25 = 2.625$.
(Exact value $8/3 = 2.6667$, so M_4 underestimates for this convex f .)
ANSWER: $M_4 = 2.625$

4.

$a_n = 2^n/n!$. Ratio:
 $a_{n+1}/a_n = [2^{n+1}/(n+1)!] \cdot [n!/2^n] = 2/(n+1)$.
 $L = \lim_{n \rightarrow \infty} 2/(n+1) = 0$.
Since $L = 0 < 1$, the series converges by the Ratio Test.
ANSWER: converges (ratio test, $L = 0 < 1$)

5.

A logarithm times another factor suggests parts, but the factor $1/x$ is precisely the derivative of $\ln x$, so substitution is far cleaner.
Let $u = \ln x$, so $du = (1/x) \, dx$.
The integral becomes $\int u \, du = u^2/2 + C$.
Back-substitute $u = \ln x$: $(\ln x)^2 / 2 + C$.
Check: $d/dx [(\ln x)^2/2] = 2(\ln x)(1/x)/2 = (\ln x)/x$.
ANSWER: $(\ln x)^2 / 2 + C$

6.

Rewrite as a limit: $\int_1^\infty x^{-2} \, dx = \lim_{t \rightarrow \infty} \int_1^t x^{-2} \, dx$.
 $\int_1^t x^{-2} \, dx = [-1/x]_1^t = -1/t - (-1) = 1 - 1/t$.
As $t \rightarrow \infty$, $1/t \rightarrow 0$, so the limit is 1.
The integral converges to 1 (a p -integral with $p = 2 > 1$).

ANSWER: converges to 1

7.

Integration by parts with $u = x$, $dv = \cos(x) dx$.

Then $du = dx$ and $v = \sin(x)$.

$$\int x \cos x dx = x \sin x - \int \sin x dx = x \sin x - (-\cos x) + C \\ = x \sin x + \cos x + C.$$

Check: $d/dx[x \sin x + \cos x] = \sin x + x \cos x - \sin x = x \cos x$.

ANSWER: $x \sin(x) + \cos(x) + C$

8.

$a_n = n/2^n$. Form the ratio a_{n+1}/a_n :

$$a_{n+1}/a_n = [(n+1)/2^{n+1}] * [2^n/n] = (n+1)/(2n).$$

$$L = \lim_{n \rightarrow \infty} (n+1)/(2n) = 1/2.$$

Since $L = 1/2 < 1$, the Ratio Test gives convergence.

ANSWER: converges (ratio test, $L = 1/2 < 1$)

9.

Alternating Series Test: $b_n = 1/n$ is positive, decreasing, and

$\lim b_n = 0$, so the series converges.

Absolute convergence? $\sum |(-1)^{n+1}/n| = \sum 1/n$ is the harmonic series, which diverges.

The series converges but not absolutely, so it is conditionally convergent.

ANSWER: conditionally convergent

10.

There is a single factor, so a substitution is tempting, but $\ln x$ has no elementary substitution antiderivative. Treat it as a product $\ln(x) \cdot 1$ and use parts.

Let $u = \ln x$ and $dv = dx$. Then $du = (1/x) dx$ and $v = x$.

$$\int \ln x dx = x \ln x - \int x(1/x) dx = x \ln x - \int 1 dx \\ = x \ln x - x + C.$$

Check: $d/dx[x \ln x - x] = \ln x + 1 - 1 = \ln x$.

ANSWER: $x \ln(x) - x + C$